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import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import norm
from scipy.stats import ks_2samp

X = [3,4,5,6,7]
Y = [4.8,4.9,5.0,5.1,5.2]

meanx = np.average(X)
stdx = np.var(X) ** 0.5

meany = np.average(Y)
stdy = np.var(Y) ** 0.5

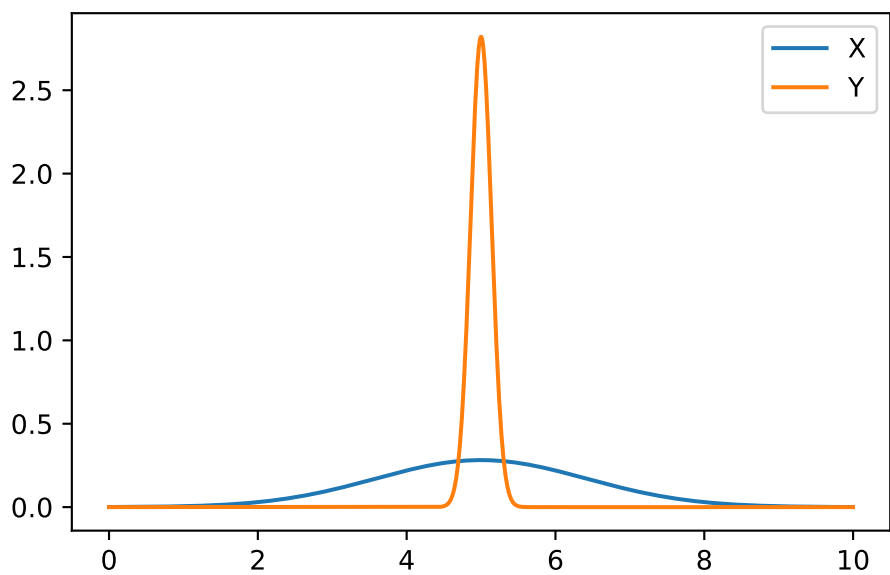
x = np.arange(0, 10, 0.001)

plt.plot(x, norm.pdf(x, meanx, stdx), label="X")
plt.plot(x, norm.pdf(x, meany, stdy), label= "Y")

plt.legend()
plt.show()

x_values = np.linspace(meanx - 4*stdx, meanx + 4*stdx, 1000)
cdf_values = norm.cdf(x_values, meanx, stdx)
plt.plot(x_values, cdf_values, label="X")
cdfy_values = norm.cdf(x_values, meany, stdy)
plt.plot(x_values, cdfy_values, label="Y")
plt.title('Normal Distribution CDF')
plt.xlabel('Value (x)')
plt.ylabel('Cumulative Probability  $P(X \leq x)$ ')
plt.grid(True)
plt.legend()
plt.show()

```



Normal Distribution CDF

